A Comparative Analysis of Two Capacitor-Less LDO Topologies in 130nm CMOS Technology

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Need for Low Dropout Regulators

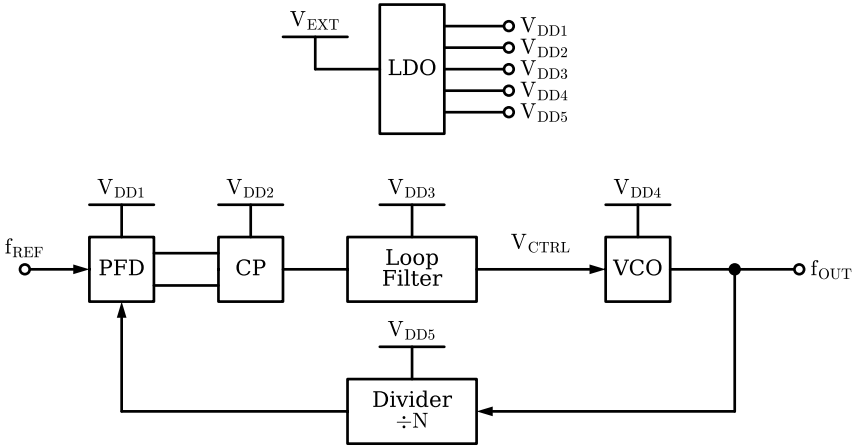


Figure: Phase Locked Loop with LDO

Target Specifications

Parameter	Target Value
Power Supply Voltage (V_{EXT})	3 V
Bias Current (I_{bias})	5 µA
Load Current (I_L)	500 µA
Load Capacitance (C_L)	50 pF
Reference Voltage (V_{REF})	$1.2\,\text{V} \pm 1\,\%$
Output Voltages (V_{OUT})	2.3 V, 2.1 V, 1.6 V
Compensation Capacitor (C_c)	< 7 pF
Phase Margin @ No-Load	$> 60^\circ$
Phase Margin @ Full-Load	$> 70^\circ$
Loop Gain @ 1 Hz	> 50 dB
PSRR @ 1 Hz	> 65 dB
PSRR @ 1 MHz	> 2 dB
PSRR @ 10 MHz	> 2 dB
Output-Referred Noise @ 10 Hz	$< -115\,\text{dBV}/\sqrt{\text{Hz}}$
Output-Referred Noise @ 1 kHz	$< -125\,\text{dBV}/\sqrt{\text{Hz}}$
Output-Referred Noise @ 10 kHz	$< -134\,\text{dBV}/\sqrt{\text{Hz}}$
Output-Referred Noise @ 100 kHz	$< -143\,\text{dBV}/\sqrt{\text{Hz}}$
Output-Referred Noise @ 1 MHz	$< -155\,\text{dBV}/\sqrt{\text{Hz}}$

Table: Technical LDO Target Specifications

PMOS LDO Schematic

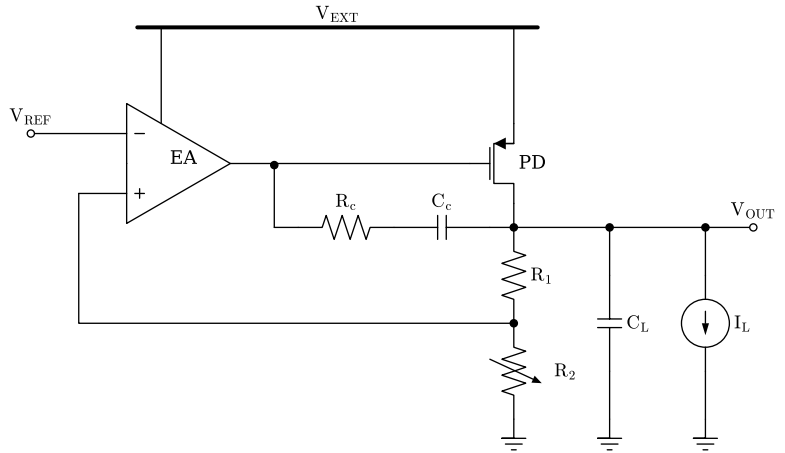


Figure: General LDO schematic

LDO's Small-Signal Model

Output voltage

$$\frac{V_{OUT}}{V_{REF}} \approx \frac{1}{\beta}, \qquad \frac{R_1}{R_2} = \frac{V_{REF}}{V_{OUT}} - 1$$

Stability

$$\omega_{p1} = \frac{1}{g_{m,PD} R_1 R_2 C_c}, \quad \omega_{p2} = \frac{g_{m,PD}}{C_1 + C_2}, \quad \omega_z = \frac{1}{C_c(1/g_{m,PD} - R_c)}$$

PSRR

$$|PSRR_{DC}| \cong \frac{1}{\beta A_{EA,0} g_{m,PD} r_{ds,p}}$$

Output-referred noise

$$S_{n,o}(f) \cong (S_{n,ref}(f) + S_{n,EA}(f))(1 + \frac{R_1}{R_2})^2 + S_{n,R2}(f)(\frac{R_1}{R_2})^2 + S_{n,R_1}(f)$$

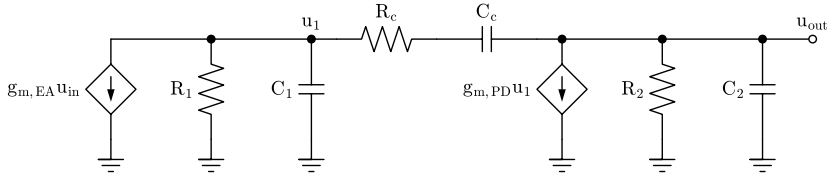


Figure: Small-signal model of the LDO

5T OTA (Simple Differential Pair)

5T OTA is a simple differential pair with a gain of:

$$A_{5T} = \frac{g_{m,12}}{g_{5T,out}}$$

whereas the output impedance:

$$g_{5T,out} = g_{ds,2} + g_{ds,4}$$

as for the output swing, in order to keep all the transistors in saturation, we must have:

$$V_{REF} - V_T \leq V_{OUT} \leq V_{EXT} - V_{DSAT}$$

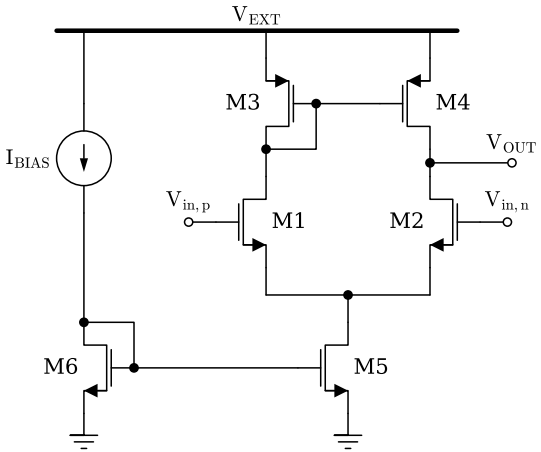


Figure: 5T OTA

5T OTA-Based LDO (Sizing)

Device	Value / Sizing	Unit
M_1, M_2 (Input Pair)	42.0/5.0	µm
M_3, M_4 (Active Load)	3.8/5.0	µm
M_5 (Tail Current)	12.0/5.0	µm
M_6 (Input Current)	5.0/5.0	µm
M_{PD} (Pass Device)	20.0/0.14	µm
R_1 (Divider)	15.00	kΩ
R_{21} (Divider)	16.36	kΩ
R_{22} (Divider)	20.00	kΩ
R_{23} (Divider)	45.00	kΩ
R_c (Compensation)	15.00	kΩ
C_c (Compensation)	7.00	pF

Table: Device Sizing for the 5T OTA LDO

- **Input Pair:** Large W and L for reducing the Flicker noise and increasing the loop gain.
- **Active Load:** Large L for increasing the loop gain, small W for improving PSRR.
- **Tail Current:** Large L for better current mirroring, large W for increasing the bias current and reducing the Thermal noise at higher frequencies.
- **Pass Device:** Large W to improve transconductance and loop gain.
- **Divider:** Small resistors to minimize noise and decrease PD's gate voltage @ No-Load.
- **Compensation:** Nulling resistor and compensation capacitor for improving UGBW and PSRR at higher frequencies.

5T OTA-Based LDO (Simulation Results)

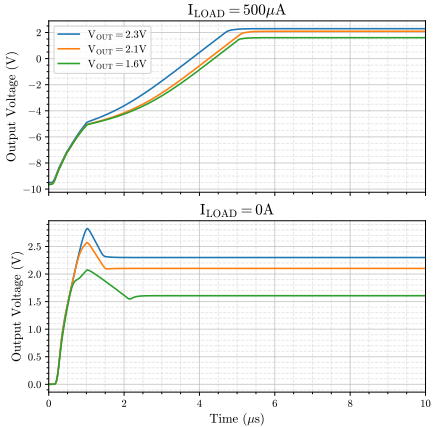


Figure: Transient Response

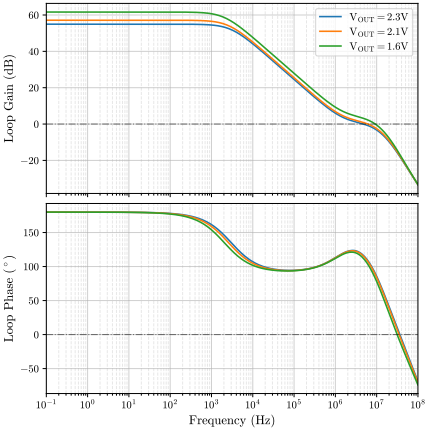


Figure: Bode Plot

5T OTA-Based LDO (Simulation Results)

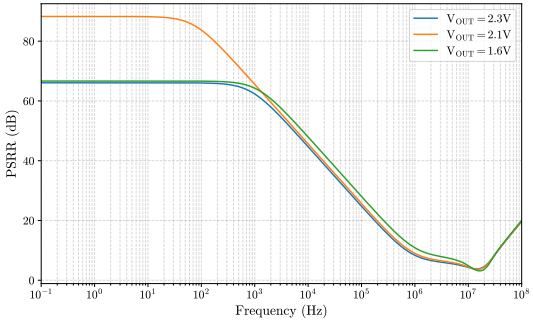


Figure: PSRR

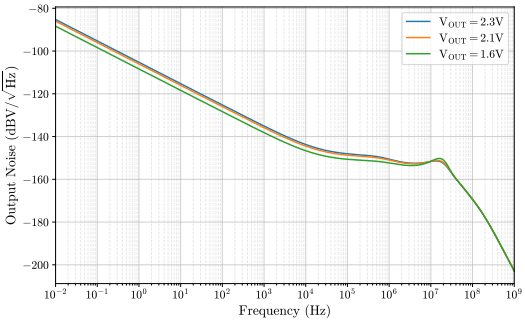


Figure: Output-Referred Noise

Telescopic OTA

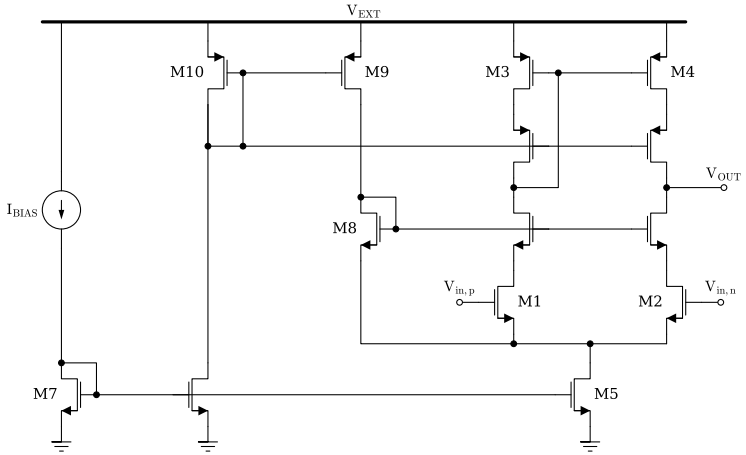


Figure: Telescopic OTA

Telescopic OTA

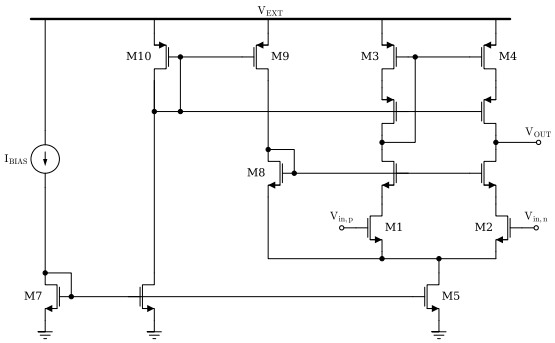


Figure: Telescopic OTA

Leading to the following output swing:

$$V_{REF} - V_T + V_{DSAT} \leq V_{OUT} \leq V_{EXT} - 2V_{DSAT}$$

Both loop gain and output impedance are substantially increased compared to the 5T OTA. More specifically, the loop gain can be given by:

$$A_{TEL} = \frac{g_{m,12}}{g_{tel,out}}$$

whereas the output impedance is given by:

$$g_{tel,out} = g_{ds,2}\frac{g_{ds,2C}}{g_{m,2C}} + g_{ds,4}\frac{g_{ds,4C}}{g_{m,4C}}$$

Bias transistors should be sized in the following manner in order to ensure maximum output swing:

$$\frac{W_8}{L_8} = \frac{W_{3,4}}{4L_{3,4}}, \quad \frac{W_{10}}{L_{10}} = \frac{W_{1,2}}{4L_{1,2}}$$

Telescopic OTA-Based LDO (Sizing)

- **Input Pair & Active Load:** Large W and L for reducing the Flicker noise and increasing the loop gain.
- **Cascode Transistors:** Small W to ensure saturation, no need for larger transconductance.
- **Current Mirror:** Large L for better current mirroring, small W for keeping all devices in saturation while improving PSRR.
- **Pass Device:** Large W to improve transconductance and loop gain.
- **Divider:** Small resistors to minimize noise and decrease PD's gate voltage @ No-Load.
- **Compensation:** Nulling resistor and compensation capacitor for improving UGBW and PSRR at higher frequencies.

Device	Value / Sizing	Unit
M_1, M_2 (Input Pair)	25.0/5.0	µm
M_{1C}, M_{2C} (NMOS Cascode)	0.65/1.0	µm
M_{3C}, M_{4C} (PMOS Cascode)	2.0/1.0	µm
M_3, M_4 (Active Load)	40.0/5.0	µm
M_8 (NMOS Bias)	0.65/4.0	µm
M_9, M_{10} (PMOS Bias)	2.0/4.0	µm
M_5 (Tail Current)	0.95/1.0	µm
M_6 (Bias Current)	0.28/1.0	µm
M_7 (Input Current)	0.28/1.0	µm
M_{PD} (Pass Device)	40.0/0.13	µm
R_1 (Divider)	22.00	kΩ
R_{21} (Divider)	24.00	kΩ
R_{22} (Divider)	29.33	kΩ
R_{23} (Divider)	66.00	kΩ
R_c (Compensation)	20.00	kΩ
C_c (Compensation)	7.00	pF

Table: Device Sizing for the Telescopic OTA LDO

Telescopic OTA-Based LDO (Simulation Results)

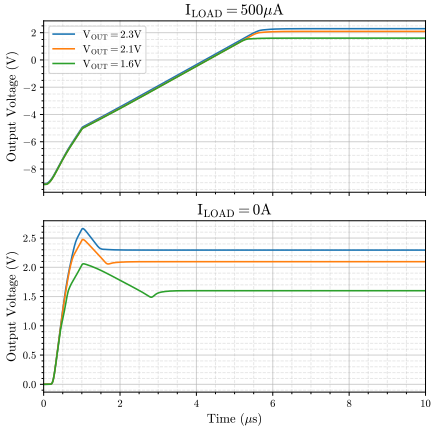


Figure: Ramp Response

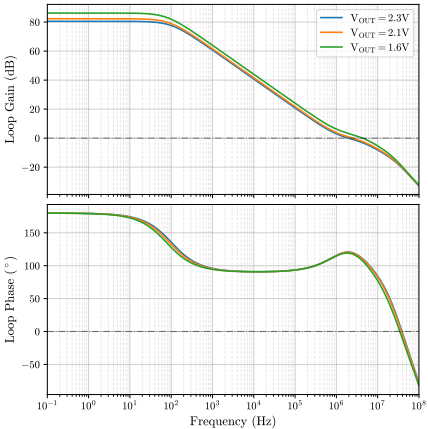


Figure: Bode Plot

Telescopic OTA-Based LDO (Simulation Results)

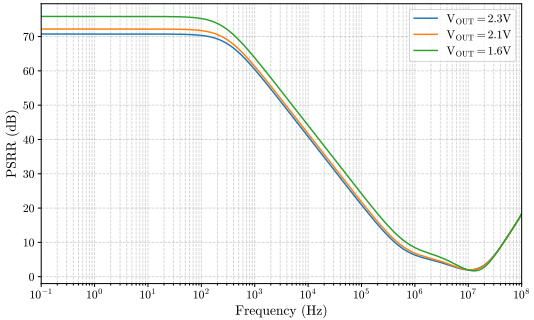


Figure: PSRR

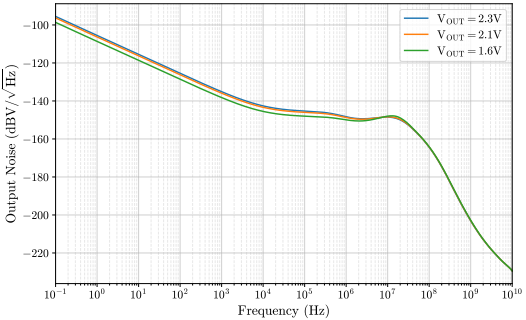


Figure: Output-Referred Noise

Simulation Results of Both Topologies

Parameter	Target Specification	5T OTA (Min – Max)	Telescopic-OTA (Min – Max)
Trim Voltage 1	$= 2.3\,\text{V}$	2.293 – 2.299	2.294
Trim Voltage 2	$= 2.1\,\text{V}$	2.096 – 2.102	2.096
Trim Voltage 3	$= 1.6\,\text{V}$	1.602 – 1.607	1.6
Quiescent Current	$< 100\,\mu\text{A}$	48.16 – 94.49	52.56 – 84.12
Phase Margin ($I_{load} = 0$)	$> 60^\circ$	77.62 – 99.2	74.15 – 99
Phase Margin ($I_{load} = 500\,\mu\text{A}$)	$> 70^\circ$	82.7 – 116.2	107.2 – 120.1
Loop Gain @ 1 Hz	$> 50\,\text{dB}$	54.95 – 61.63	80.53 – 86.23
PSRR @ 1 Hz	$> 65\,\text{dB}$	66.07 – 88.27	70.73 – 75.86
PSRR @ 1 MHz	$> 2\,\text{dB}$	8.43 – 10.95	6.373 – 8.537
PSRR @ 10 MHz	$> 2\,\text{dB}$	4.353 – 4.934	2.107 – 2.118
Output-Referred Noise @ 10 Hz	$< -115\,\text{dBV}/\sqrt{\text{Hz}}$	-118.3 – -115.1	-118.6 – -115.5
Output-Referred Noise @ 1 kHz	$< -125\,\text{dBV}/\sqrt{\text{Hz}}$	-138.1 – -135	-138.1 – -135.1
Output-Referred Noise @ 10 kHz	$< -134\,\text{dBV}/\sqrt{\text{Hz}}$	-146.5 – -143.6	-145.4 – -142.5
Output-Referred Noise @ 100 kHz	$< -143\,\text{dBV}/\sqrt{\text{Hz}}$	-150.6 – -148	-148 – -145.3
Output-Referred Noise @ 1 MHz	$< -155\,\text{dBV}/\sqrt{\text{Hz}}$	-152.3 – -150.5	-149.9 – -148.3

Table: Simulated Performance Summary (Across All Trims)

Comparison

- **5T OTA-Based LDO:** Offers the most balanced results. Should be preferred when:
  1. Smaller Pass Device is needed.
  2. Better EA output swing is required for keeping all devices in saturation.
  3. Searching for a simpler design with fewer transistors.
  4. Need for slightly better noise results @ high frequencies.
  5. Need for slightly better PSRR results @ high frequencies.
- **Telescopic OTA-Based LDO:** Maximizes loop gain and PSRR. Should be preferred when:
  1. Larger Pass Device can be utilized.
  2. No care for the limited output swing of the EA.
  3. Can work on more complex designs with more transistors.
  4. Need for higher PSRR values @ low frequencies.
  5. Need for better Phase Margin @ Load.

Conclusion and Future Work

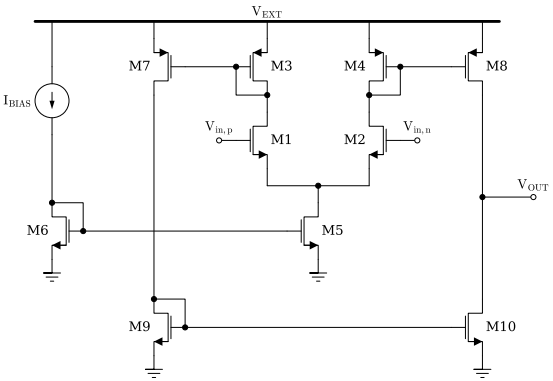


Figure: Current Mirror OTA

Both designs performed well across almost all proposed specifications and failed only to meet the output-referred noise specification at higher frequencies.Achieving the noise specification at higher frequencies requires increasing the input-transistor bias current, as thermal noise dominates in this regime. However, increasing the bias current in both topologies reduces the output impedance, thereby decreasing loop gain and PSRR performance. Alternative architectures with separate input and gain stages, each optimized for different operating currents, warrant investigation.

Current Mirror OTA

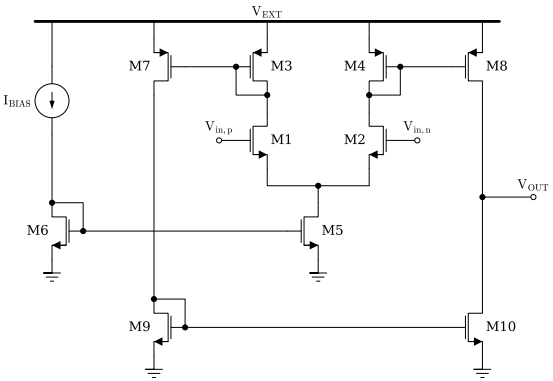


Figure: Current Mirror OTA

Loop gain can be improved by mirroring factor B:

$$A_{CM} = \frac{B g_{m,12}}{g_{cm,out}}$$

whereas the output impedance is given by:

$$g_{cm,out} = g_{ds,8} + g_{ds,10}$$

Current Mirror OTA-Based LDO (Sizing)

Device	Value / Sizing	Unit
M_1, M_2 (Input Pair)	40.0/0.5	µm
M_3, M_4 (Active Load)	5.0/5.0	µm
M_7, M_8 (PMOS CM)	2.5/5.0	µm
M_9, M_{10} (NMOS CM)	1.0/5.0	µm
M_5 (Tail Current)	16.0/1.0	µm
M_6 (Bias Current)	8.0/1.0	µm
M_{PD} (Pass Device)	40.0/0.5	µm
R_1 (Divider)	22.00	kΩ
R_{21} (Divider)	24.00	kΩ
R_{22} (Divider)	29.33	kΩ
R_{23} (Divider)	66.00	kΩ
R_c (Compensation)	15.00	kΩ
C_c (Compensation)	7.00	pF

Table: Device Sizing for the Current Mirror OTA LDO

Current Mirror OTA-Based LDO (Simulation Results)

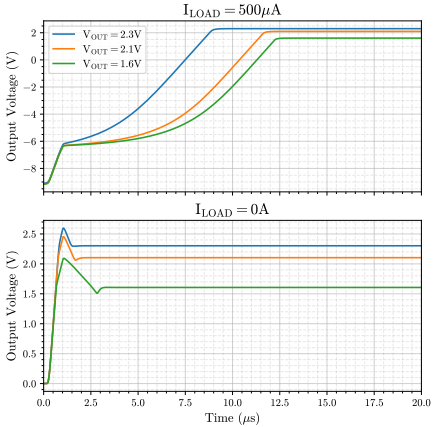


Figure: Ramp Response

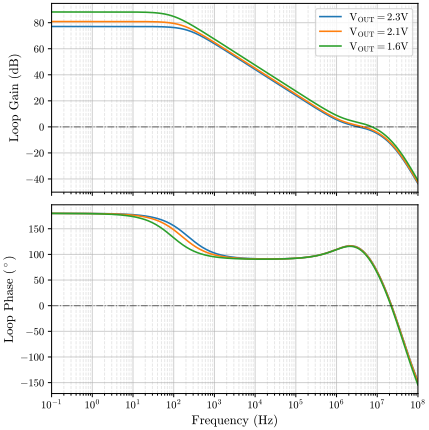


Figure: Bode Plot

Current Mirror OTA-Based LDO (Simulation Results)

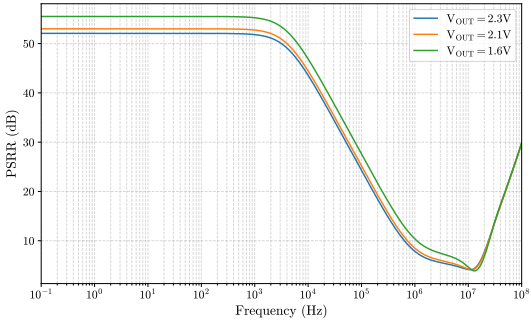


Figure: PSRR

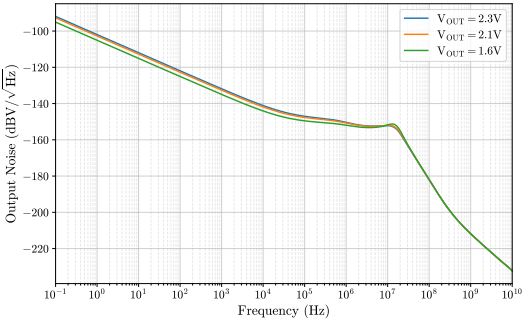


Figure: Output-Referred Noise

Current Mirror OTA-Based LDO Simulation Results

Parameter	Target Specification	Current Mirror OTA (Min – Max)
Trim Voltage 1	$= 2.3\,\text{V}$	2.301 – 2.303
Trim Voltage 2	$= 2.1\,\text{V}$	2.103 – 2.104
Trim Voltage 3	$= 1.6\,\text{V}$	1.605 – 1.606
Quiescent Current	$< 100\,\mu\text{A}$	54.26 – 85.94
Phase Margin ($I_{load} = 0$)	$> 60^\circ$	64.39 – 85.28
Phase Margin ($I_{load} = 500\,\mu\text{A}$)	$> 70^\circ$	82.14 – 112.3
Loop Gain @ 1 Hz	$> 50\,\text{dB}$	77.09 – 88.23
PSRR @ 1 Hz	$> 65\,\text{dB}$	52.07 – 55.5
PSRR @ 1 MHz	$> 2\,\text{dB}$	7.923 – 10.45
PSRR @ 10 MHz	$> 2\,\text{dB}$	4.337 – 4.802
Output-Referred Noise @ 10 Hz	$< -115\,\text{dBV}/\sqrt{\text{Hz}}$	-111.9 – -115
Output-Referred Noise @ 1 kHz	$< -125\,\text{dBV}/\sqrt{\text{Hz}}$	-131.8 – -134.9
Output-Referred Noise @ 10 kHz	$< -134\,\text{dBV}/\sqrt{\text{Hz}}$	-141 – -144
Output-Referred Noise @ 100 kHz	$< -143\,\text{dBV}/\sqrt{\text{Hz}}$	-147 – -149.5
Output-Referred Noise @ 1 MHz	$< -155\,\text{dBV}/\sqrt{\text{Hz}}$	-150.3 – -151.9

Table: Simulated Performance Summary (Across All Trims)

Acknowledgements

This work wouldn't have been possible without the help of PhD candidate Andreas Papathanasiou and Pr. Paul-Peter Sotiriadis. Special thanks go to Dr. Dimitrios Baxevanakis for his guidance and help all throughout the project.

Thank You!
Any Questions?
el22174@mail.ntua.gr

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